

NATIONAL OCCUPATIONAL STANDARD FOR BIOMEDICAL SCIENTIST



**NOS.BS.01
FIRST EDITION**

APPROVING AUTHORITY

This National Occupational Standard (NOS) has been prepared and published under the authority of the Zambia Qualifications Authority Board on 20th December, 2023.

ZAMBIA QUALIFICATIONS AUTHORITY

The Zambia Qualifications Authority (ZAQA) Act No. 13 of 2011 was enacted by the Government of the Republic of Zambia to ***“provide for the development and implementation of a national qualifications framework; establish the Zambia Qualifications Authority; provide for the registration and accreditation of qualifications; provide measures to ensure that standards and registered qualifications are internationally comparable; and provide for matters connected with, or incidental to the foregoing”***. Among other functions, ZAQA is responsible for ***determining national standards for any occupation***, through the various sector specific National Occupational Standards Development Teams (NOSDTs).

REVISION OF NATIONAL OCCUPATIONAL STANDARDS

National Occupational Standards (NOS) shall be revised every 5 years, or whenever it is deemed necessary, by the issuance of either amendments or of revised editions. It is important that users of the NOS ascertain that they are in possession of the latest amendments or editions.

NOS DEVELOPMENT TEAM RESPONSIBLE

This National Occupational Standard was prepared by the Manufacturing NOSDT, upon which the following organisations were represented:

1. Biomedical Society of Zambia (BMSZ)
2. Copperbelt University (CBU)
3. Engineering Institution of Zambia (EIZ)
4. Good Time Steel Company Zambia Limited (GTS)
5. Lusaka Vocational and Technical College (LVTC)
6. Makeni Ecumenical Centre (MEC)
7. Ministry of Commerce, Trade and Industry (MCTI)
8. Ministry of Health (MoH)
9. National Institute for Scientific and Industrial Research (NISIR)
10. Northern Technical College (NORTEC)
11. University of Lusaka (UNILUS)
12. University of Zambia (UNZA)
13. Zambia Association of Manufacturers (ZAM)
14. Zambia Forestry College (ZFC)
15. Zambia Qualifications Authority (ZAQA) – Secretariat

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FOREWORD

The Zambia Qualifications Authority (ZAQA) is a statutory body under the Ministry of Education established by ZAQA Act No. 13 of 2011 to ***“develop and implement a national qualifications framework; register and accredit qualifications; and ensure that standards and registered qualifications are internationally comparable.”***

Among other functions, ZAQA is responsible for ***“determining national standards for any occupation”***, through the various sector specific National Occupational Standards Development Teams (NOSDTs) with experts composed of representation from the appropriate authorities, government departments, consumer associations, regulators, industry, academia, and non-governmental organisations, etc.

This National Occupational Standard (NOS) has been developed by the Manufacturing National Occupational Standards Development Team in accordance with the laid down procedures and guidelines of ZAQA. All users should ensure that they have the latest edition of this publication as the National Occupational Standards are revised from time to time.

This NOS shall be used by, among others, industry, employers, quality assurance bodies, awarding and professional bodies, and education and training institutions, as a benchmark to identify training needs, develop job profiles/descriptions, develop curricula, and learning programmes in various sectors where the occupation exists. In the Manufacturing sector, demonstration of competence against this NOS may be required in order to run a business or practice a craft or profession.

JUSTIFICATION

Over the last 20 to 30 years, Medicine and Health practice has become complex due to the emergence of new diseases and technologies that are used to diagnose them. Biomedical Sciences have evolved to provide specialised innovative methods and technologies for detecting and monitoring the new health threats, such as Covid-19, while improving the detection and monitoring of old diseases, such as Malaria. As such, a Biomedical Scientist is cardinal, while being central, in healthcare provision by performing laboratory in-vitro diagnostic tests through the application of skills and knowledge in core competences of haematology and blood transfusion science, medical microbiology, histopathology, clinical chemistry, medical parasitology, virology, and molecular biology

Overall, the work of Biomedical Scientists is a vital linkage between clinical care, nursing, and therapy. A Biomedical Scientist supports the work of other healthcare professionals such as medical doctors, nurses, pharmacists, and health policy makers through medical research. This National Occupational Standard highlights the core knowledge, skills, competences and personal attributes that Biomedical Scientists must possess to be successful in their jobs.

ACRONYMS AND ABBREVIATIONS

BS	Biomedical Scientist
CS	Core Skills
FEFO	First Expire, First Out
K	Knowledge and Understanding
LIMS	Laboratory Information Management System
NOS	National Occupational Standard
NOSDT	National Occupational Standards Development Team
OK	Organisational Context
OSHE	Occupational Safety, Health and Environment
PC	Performance Criteria
PS	Professional Skills
RK	Regulatory Context
RPL	Recognition of Prior Learning
S	Skills
TK	Technical Knowledge
ZAMMSA	Zambia Medicines and Medical Supplies Agency
ZAQA	Zambia Qualifications Authority
ZQF	Zambia Qualifications Framework

GLOSSARY OF TERMS

For the purposes of this NOS, the following terms and definitions shall apply:

Core Skills/Generic Skills: are a group of skills that are key to learning and working in today's world. These skills are typically needed in any work environment. In the context of the NOS, these include communication related skills that are applicable to most job roles.

Disability: is the physical or mental impairment that substantially limits one or more major life activities.

Function: is an activity necessary for achieving the key purpose of the sector, occupation or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of the NOS.

Job Title: defines a unique set of functions that together form a unique employment opportunity in an organisation.

Knowledge and Understanding: are statements which together specify the technical, generic, professional, and organisational specific knowledge that an individual needs in order to perform to the required standard.

National Occupational Standards (NOS): are statements of the standards of performance individuals must achieve when carrying out functions in the workplace, together with specifications of the underpinning knowledge and understanding. They are precise descriptions of what an individual is expected to be able to do in his/her work role.

National Occupational Standards (NOS) Code: is a unique reference code that identifies a NOS.

National Occupational Standards Development Team (NOSDT): means an established group of national stakeholders/experts responsible for the development of National Occupational Standards within a specific economic sector or occupation.

Occupation: is a set of job roles which perform similar/related set of functions in an industry.

Organisational Context: includes the way the organisation is structured and how it operates, including the extent of operative knowledge that managers have in their relevant areas of responsibility.

Performance Criteria: are statements that together specify the standard of performance required when carrying out a task.

Scope: is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.

Sector: is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy, whose components share similar characteristics and interests.

Sub Sector: is derived from a further breakdown based on the characteristics and interests of its components.

Technical Knowledge: is the specific knowledge needed to accomplish specific designated responsibilities.

Unit Title: gives a clear overall statement about what the incumbent should be able to do.

Elements: set out competences the incumbent should possess to carry out the day-to-day activities.

1. OVERVIEW

This is an introductory section providing a summary and specific information or commentary about the content of the NOS, and the targeted sector and occupation to help the user judge whether it is relevant to them.

NOS Code	NOS.BS.01
Occupation	Biomedical Sciences
Job Title	Biomedical Scientist
Job Description	A Biomedical Scientist is critical in the provision of primary and secondary healthcare services. He/she is responsible for the supervision of subordinates, coordination of laboratory work, validation of test methods and results, monitoring and training of staff and students on new methods and procedures, evaluation of quality assurance processes, safety in the medical laboratory, and conducting medical research to contribute to advancement in medical technology.
Job Purpose	Biomedical Scientists perform a wide range of medical diagnostic tests, conduct and publish biomedical research findings, and offer teaching and training in learning institutions.
ZQF Level	7
Sector	Manufacturing
Sub sectors	Health, Education, Regulatory Bodies, e.g. HPCZ, ZAMRA, NHIMA, ZAMMSA.
Other Economic Sector(s) in which the Occupation is Practiced	Water and Sanitation, Government Ministries and Agencies, Food and Drug, Mining (Health & Safety), etc.
Other Similar Jobs that can be performed in the Occupation	Tutor/Lecturer/Trainer/Researcher, Biomedical Sales Manager, Consultant, Assistant Director – Laboratory Services (MoH), etc.
Minimum Educational Job Entry Qualification	Bachelor's Degree in Biomedical Sciences
Practicing License Requirements (if any)	Membership with the Health Professions Council of Zambia (HPCZ)
Training/RPL	1. A-Levels: Entry Year 2 of Training 2. Diploma in Biomedical Sciences: Entry Year 3 of Training 3. Any Health-Related Degree: Entry Year 2 of Training 4. Use of ICT
Minimum Job Entry Age	20 years
Prior Experience (Suggested)	Minimum of 1 Year Internship
Performance Criteria	As described in the Units under Section 4

2. SCOPE

This NOS highlights the core competences, skills, knowledge and understanding, and personal attributes that Biomedical Scientists are required to possess to be successful in their jobs.

3. PERSONAL ATTRIBUTES (VALUES, ETHICS AND ATTITUDES)

The Biomedical Scientist should possess the following attributes:

- Sound research skills;
- Attention to detail;
- Problem-solving skills;
- Analytical skills;
- High level of professional and ethical practice standards;
- Strong interpersonal skills;
- Be polite, considerate, trustworthy, and honest;
- Act with integrity, maintain confidentiality, respect patients' dignity and privacy.

4. UNITS AND ELEMENTS

This National Occupational Standard is divided into 8 Units, representing the tasks that a jobholder should undertake in his/her day-to-day work. Each unit is further broken down into elements depicting the number of activities to be carried out for the successful execution of a particular task.

UNIT 1 [This Unit covers the skills and knowledge required by a Biomedical Scientist in demonstrating laboratory safety and waste management]

Unit No.	01
Unit Title	Medical Laboratory Safety and Waste Management
Description	This Unit describes the skills and knowledge required by a Biomedical Scientist to safely work and dispose of waste materials in a medical laboratory,
Scope	This Unit covers the following: • Use of Personal Protective Equipment (PPE) and First Aid Box. • Awareness of the Types of Hazards. • Types of Medical Laboratory Waste. • Safe Waste Disposal. • Disinfection, Decontamination, and Incineration.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Use of Personal Protective Equipment (PPE) and First Aid Box	To be competent, the individual must be able to: PC1. Use PPE appropriately; - Wear gloves, lab coat, pressurised suit, safety goggles - Wear appropriate closed shoes - Tie back long hair PC2. Use the First Aid Box appropriately.
Awareness of the Types of Hazards	To be competent, the individual must be able to: PC3. Identify types of hazards in the laboratory; - Fire hazards - Bio-hazards - Chemical hazards - Electrical hazards
Types of Medical Laboratory Waste	To be competent, the individual must be able to: PC4. Recognise types of laboratory waste; - Biological waste - Non-biological waste - Sharp waste
Safe Waste Disposal	To be competent, the individual must be able to: PC5. Dispose of laboratory waste safely. PC6. Dispose waste according to the Laboratory Safety Manual.
Disinfection, Decontamination, and Incineration	To be competent, the individual must be able to: PC7. Disinfect work surfaces in the Lab before beginning work according to the Laboratory Safety Manual. PC8. Autoclave reusable laboratory instruments. PC9. Autoclave infectious biological wastes and incinerate sharps.

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the laboratory safety manual and its guidelines)	The individual on the job must demonstrate knowledge and understanding of: OK1. Laboratory Safety Manual procedures. OK2. Good Laboratory Practice Manual. OK3. Standard Operating Procedures (SOP).
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. Risk assessment. TK2. Disposal of PPEs. TK3. Contents and use of the First Aid Box. TK4. Different colour codes of bin liners for waste disposal. TK5. Types of biological and non-biological waste. TK6. Use and preparation of disinfectants according to the SOP. TK7. Use and operation of the different types of safety cabinets. TK8. Operations of an Autoclave.
C. Regulatory Knowledge (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Relevant Occupational Health and Safety laws. RK2. Relevant Zambia National Biomedical Laboratory Safety Manual. RK3. Relevant World Health Organisation Laboratory Safety Manual. RK4. Zambia Environmental Management Agency, (Act 2011).
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English language proficiently and/or have the means to write complex instructions in reports and SOP. CS2. Prepare and provide clear and detailed instructions, and details of pathways and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English language proficiently, and be able to give detailed instructions. CS4. Read and interpret complex sketches, operating manual drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers. CS6. Convey information clearly, concisely, and proficiently to co-workers.

B. Professional Skills	Decision-Making
	The individual on the job must be able to:
	PS1. Decide whether the laboratory is safe for working, and ensure that a particular task is not creating hazardous conditions for other workers.
	PS2. Decide whether the equipment setup is safe for the intended work flow.
	Plan and Organise
	The individual on the job should be able to:
	PS3. Decide if the laboratory work space is sufficient for its intended use.
	PS4. Appoint a Laboratory Safety Officer to enforce health standards according to the Safety Manual.
	PS5. Develop working rotas and work allocation.
	PS6. Allocate sufficient resources for procuring safety equipment and maintenance of safety in the laboratory.
	Customer Centricity
	The individual on the job should be able to:
	PS7. Manage relationships with patients and other clients by ensuring professionalism and confidentiality.
	Problem-Solving
	The individual on the job should be able to:
	PS8. Resolve any conflicts with the subordinates and colleagues professionally, and where necessary, refer to HR department.
	Analytical Thinking
	The individual on the job should be able to:
	PS9. Analyse and categorise potentially different safety challenges within the laboratory.
	Critical Thinking
	The individual on the job should be able to:
	PS10. Identify the common safety lapses, and the types of accidents that occur in the laboratory.

UNIT 2 [This Unit covers the skills and knowledge required by a Biomedical scientist in demonstrating practical medical laboratory skills to perform diagnostic procedures]

Unit No.	02
Unit Title	Practical Medical Laboratory Skills to Perform Diagnostic Procedures
Description	This Unit describes the skills and knowledge required by a Biomedical Scientist to carry out laboratory diagnostic procedures in haematology and blood transfusion science, clinical chemistry, medical microbiology, medical parasitology, medical virology, and molecular biology; phlebotomy and specimen collection; histopathology and immunology departments in the medical laboratory.
Scope	This Unit covers the following: • Application of Practical Skills When Performing Diagnostic Procedures. • Detailed Analytical Skills to Produce Accurate and Reliable Results. • Attention to Detail When Performing Diagnostic Procedures. • Analyse and Interpret Data Produced After Conducting the Diagnostic Procedure. • Conduct Quality Assurance Tests Before Using the Equipment for Diagnostic Procedures.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Application of Practical Skills When Performing Diagnostic Procedures	To be competent, the individual must be able to: PC1. Apply medical laboratory skills. PC2. Demonstrate practical skills developed during training. PC3. Perform diagnostic procedures according to SOP.
Detailed Analytical Skills to Produce Accurate and Reliable Results	To be competent, the individual must be able to: PC4. Analyse information related to the specific procedure. PC5. Identify additional requirements for the specific procedures. PC6. Apply problem-solving skills. PC7. Think through the task critically. PC8. Brainstorm with other scientists to achieve accurate results.
Attention to Detail When Performing Diagnostic Procedures	To be competent, the individual must be able to: PC9. Perform laboratory diagnostic procedures according to SOP. PC10. Organise workspace, reagents, and equipment before commencing the laboratory diagnostic procedure. PC11. Carry out the laboratory diagnostic techniques accurately PC12. Conduct the laboratory diagnostic techniques according to routine procedures. PC13. Concentrate throughout the process of the procedure.

Analyse and Interpret Data Produced After Conducting the Diagnostic Procedure	To be competent, the individual must be able to: PC14. Analyse each component of the data to arrive at the accurate laboratory diagnosis. PC15. Explain scientifically what the findings of the procedure mean. PC16. Present, select, organise, and group ideas and evidence in a logical manner on the Laboratory Request Form.
Conduct Quality Assurance Tests Before Using the Equipment for Diagnostic Procedures	To be competent, the individual must be able to: PC17. Conduct quality assurance tests on equipment before testing patients' samples. PC18. Schedule service time for all equipment in the medical laboratory to maintain accurate performance of the equipment. PC19. Ensure that the Quality Assurance Officer enforces quality assurance management issues in the medical laboratory.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the Standard Operating Procedures and its guidelines)	The individual on the job must demonstrate knowledge and understanding of: OK 1. Standard Operating Procedures. OK 2. Good Laboratory Practice Manual. OK 3. Laboratory Safety Manual.
B. Technical Knowledge	The individual on the job must demonstrate technical knowledge and understanding of: TK1. Laboratory skills in Haematology and Blood Transfusion Science – staining of blood slides, blood typing, and cross-matching. TK2. Laboratory skills in Medical Microbiology – microscopy, culture and sensitivity testing skills, and aseptic technique skills. TK3. Laboratory skills in Medical Parasitology – preparation of thin and thick smears, and wet preparation of slides for microscopy. TK4. Laboratory skills in Virology and Molecular Biology – PCR technique. TK5. Laboratory skills in Histopathology – tissue processing and staining. TK6. Laboratory skills in Immunology – Serology. TK7. Laboratory skills in Phlebotomy using the syringe or Vacutainer methods.
C. Regulatory Knowledge (Knowledge of rules and regulations)	The individual on the job must demonstrate knowledge and understanding of: RK1. Relevant Occupational Health and Safety laws. RK2. The Zambia National Biomedical Laboratory Safety Manual.

	RK3. Relevant World Health Organisation Laboratory Safety manuals. RK4. Relevant Health Professional Acts and regulations,
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English language proficiently, and be able to write complex instructions in reports and SOP. CS2. Prepare and provide clear and detailed instructions, details of pathways and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English language proficiently, and be able to give detailed instructions. CS4. Read and interpret complex sketches, operating manual drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Decide whether the equipment used for the specific diagnostic procedure is operating normally to produce accurate and reliable results. PS2. Decide whether the results produced can be well interpreted to help in the further management of the patient by the medical staff.
	Plan and Organise
	The individual on the job should be able to: PS3. Decide if the laboratory working space is sufficient for its intended use. PS4. Ensure that the Quality Assurance Officer enforces health standards regarding laboratory equipment and their performance. PS5. Allocate sufficient resources for procuring laboratory reagents and consumables required to conduct diagnostic procedures.
	Customer Centricity
	The individual on the job should be able to: PS6. Manage relationships with patients and other clients by ensuring that confidentiality is maintained.

	Problem-Solving
	The individual on the job should be able to:
	PS7. Ensure that all laboratory equipment is operating normally, and that reagents and consumables are available to conduct diagnostic procedures.
	PS8. Report to senior management if the equipment is not working normally so that arrangements for repair or ordering for new equipment are included in the plan.
	Analytical Thinking
	The individual on the job should be able to:
	PS9. Analyse and categorise potentially different challenges within the laboratory affecting the performance of diagnostic procedures.
	Critical Thinking
	The individual on the job should be able to:
	PS10. Identify the common challenges with diagnostic equipment, reagents, and consumables shortages in a medical laboratory.

UNIT 03 [This Unit covers the skills and knowledge by a Biomedical Scientist in logistics and supply chain management]

Unit No.	03
Unit Title	Logistics and Supply Chain Management
Description	This Unit describes the skills and knowledge required by a Biomedical Scientist to use in logistics and supply chain management of HIV test kits and laboratory commodities.
Scope	This Unit covers the following: • Use of Standard Operating Procedures Manual for Management of the National HIV Test Logistics System. • Use of SOP Manual for Management of the National Laboratory Commodity Logistics System.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Use of Standard Operating Procedures Manual for Management of the National HIV Test Logistics System	To be competent, the individual must be able to: PC1. Identify his/her roles and responsibilities in the National HIV Test Logistics System. PC2. Identify the forms used in managing the National HIV Test Logistics System. PC3. Employ the Job Aid in the SOP Manual to perform tasks. PC4. Operate the electronic Laboratory Information Management System (eLIMS) Central Edition or Facility Edition to place orders with ZAMMSA.
Use of SOP Manual for Management of the National Laboratory Commodity Logistics System	To be competent, the individual must be able to: PC5. Identify his/her roles and responsibilities in the SOP Manual for management of the National Laboratory Commodity Logistics System. PC6. Identify the forms used in managing the HIV Test Logistics System. PC7. Employ the Job Aid in the SOP Manual to perform tasks. PC8. Operate the eLIMS Central Edition and Facility Edition to place orders with ZAMMSA.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the SOP Manual for management of the National Laboratory Commodity Logistics System and HIV Logistic System)	The individual on the job must demonstrate knowledge and understanding of: OK1. Use of each form in the HIV test kits and the National Laboratory Commodity Logistics System. OK2. Use of the eLIMS.

B. Technical Knowledge (The National HIV Test Logistics System and National Laboratory Commodity Logistics System)	The individual on the job must demonstrate knowledge and understanding of: TK1. His/her roles and responsibilities in the Supply Chain Management of the HIV test kit system. TK2. Recording the usage of HIV test kits. TK3. Assessing stock status. TK4. Reporting and ordering HIV test kits. TK5. Receiving HIV test kits. TK6. Storing of HIV test kits. TK7. The National Laboratory Commodity Logistics System. TK8. Recording the usage of the laboratory commodities. TK9. Receiving and storage of laboratory commodities.
C. Regulatory Knowledge (Knowledge of rules and regulation)	The individual on the job must demonstrate knowledge and understanding of: RK1. The SOP Manual for management of the National Laboratory Commodity Logistics System. RK2. The SOP Manual for management of the National HIV Test Logistics System. RK3. Guidelines for managing the Laboratory Supply Chain (v.2), USAID DELIVER PROJECT
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English language proficiently, and be able to write complex instructions in reports and SOP. CS2. Prepare and provide clear and detailed instructions, details of pathways and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English proficiently, and be able to give instructions. CS4. Read and interpret LIMS document or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
B. Professional Skills	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers. CS6. Convey information clearly and concisely to co-workers.
	Decision-Making
	The individual on the job must be able to: PS1. Decide which Job Aid to use when performing an action from the SOP manuals for the management of the National Laboratory Commodity Logistics System and the National HIV Test Logistics System. PS2. Regularly access stock status for each commodity in the Supply Chain System for decision-making.

	Plan and Organise
	The individual on the job should be able to:
	PS3. Decide and ensure availability of the Supply Chain Management System.
	PS4. Use the Facility Edition or Central Edition of the eLIMS.
	Customer Centricity
	The individual on the job should be able to:
	PS5. Manage the Logistics and Supply Chain Management System for continuous service delivery.
	Problem-Solving
	The individual on the job should be able to:
	PS6. Resolve challenges in the operations of Logistics and Supply Chain Management within the laboratory.
	PS7. Contact supervisors if the challenges in the Logistics and Supply Chain Management System cannot be resolved.
	Analytical Thinking
	The individual on the job should be able to:
	PS8. Analyse challenges of the Logistics and Supply Chain Management System.
	Critical Thinking
	The individual on the job should be able to:
	PS9. Employ corrective measures to address challenges in the Logistics and Supply Chain Management System.

UNIT 04 [This Unit covers the skills and knowledge required by a Biomedical Scientist in main laboratory stores management]

Unit No.	04
Unit Title	Laboratory Stores Management
Description	This Unit describes the skills and knowledge required by a Biomedical Scientist in Main Laboratory Stores Management.
Scope	This Unit covers the following: • Securing Laboratory Storeroom. • Storage of Laboratory Commodities. • Laboratory Inventory Management.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Securing Laboratory Storeroom	To be competent, the individual must be able to: PC1. Ensure and check that the storeroom is strongly built with a secure door.
Storage of Laboratory Commodities	To be competent, the individual must be able to: PC2. Check and identify the reagents, consumables, and durables. PC3. Stack the reagents, consumables, and durables in their respective sections. PC4. Identify laboratory commodities with special storage requirements and store them accordingly. PC5. Identify and store laboratory commodities according to the principle of First Expire, First Out (FEFO). PC6. Approve that the laboratory commodities are physically Inspected.
Laboratory Inventory Management	To be competent, the individual must be able to: PC7. Put each Stock Control Card next to the specific laboratory commodity. PC8. Record every transaction on the Laboratory Stock Control after every issuance or receipt of stock. PC9. Report any theft or loss of stock of any laboratory commodity.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the SOP for the Commodity Logistics System)	The individual on the job must demonstrate knowledge and understanding of: OK1. Use of each LIMS document in the HIV test kit and the National Laboratory Commodity Logistics System. OK2. Conducting the physical count every month end.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. A wide variety of storage requirements for different commodities for their maintenance. TK2. Different reagents, consumables, and durables. TK3. Physical condition of each commodity. TK4. Principle of FEFO.

C. Regulatory Knowledge (Knowledge of rules and regulation)	The individual on the job must demonstrate knowledge and understanding of: RK1. Guidelines for managing the Laboratory Supply Chain (v.2), July 2008 RK2. SOP Manual for management of the National Laboratory Commodity Logistics System. RK3. SOP Manual for management of the National HIV Test Logistics System.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English language proficiently, and be able to write complex instructions in reports and SOP. CS2. Prepare and provide clear and detailed instructions, details of pathways and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English proficiently, and be able to give detailed instructions. CS4. Read and interpret LIMS document or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers. CS6. Convey information clearly, concisely, and proficiently to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Decide the layout of the storeroom for maximum use of storage space. PS2. Regularly conduct physical inspection on the state of the storeroom and stock status for each commodity for decision-making.
	Plan and Organise
	The individual on the job should be able to: PS3. Decide where consumables, reagents, and durables are stored in the storeroom for safety. PS4. Decide the storage of flammable, corrosive commodities requiring cold storage.
	Customer Centricity
	The individual on the job should be able to: PS5. Manage the storeroom and ensure availability of commodities for the benefit of the patients.

	Problem-Solving
	The individual on the job should be able to: PS6. Resolve challenges in the storeroom related to receiving and issuing of commodities. PS7. Request for more space for better management and storage of commodities. PS8. Appoint a dedicated Stores Manager.
	Analytical Thinking
	The individual on the job should be able to: PS9. Analyse challenges associated with poor storage conditions of commodities.
	Critical Thinking
	The individual on the job should be able to: PS10. Employ corrective measures to address challenges with shortages of laboratory commodities.

UNIT 5 [This Unit covers the skills and knowledge required by a Biomedical Scientist in demonstrating knowledge on conducting medical research and publications]

Unit No.	05
Unit Title	Medical Research and Publications
Description	This Unit describes the skills and knowledge required by a Biomedical Scientist to carry out medical research and publications.
Scope	This Unit covers the following: • Creativity. • Data Collection and Analysis. • Academic Writing and Organisation. • Time Management. • Research Budget Estimation. • Publication of Research Findings.
Element	Performance Criteria (PC)
Creativity	To be competent, the individual must be able to: PC1. Develop or recognise new ideas in biomedical sciences. PC2. Think flexibly and view scientific issues in a different perspective. PC3. Use imagination or original ideas to create something new and useful in problem-solving. PC4. Develop a proposal to communicate the new ideas and look for financial support.
Data Collection and Analysis	To be competent, the individual must be able to: PC5. Perform experiments in the laboratory. PC6. Develop survey questionnaires to collect data from the study participants. PC7. Conduct data sorting and cleaning. PC8. Analyse data either qualitatively or quantitatively or both by using the appropriate software packages.
Academic Writing and Organisation	To be competent, the individual must be able to: PC9. Use scientific writing skills and communicate information gathered clearly and concisely. PC10. Demonstrate how to convey arguments in a logical and convincing manner. PC11. Use accurate information which is clear and relevant to the research topic.
Time Management	To be competent, the individual must be able to: PC12. Use time efficiently and keep track of research deadlines to conduct the research within the stipulated period.
Research Budget Estimation	To be competent, the individual must be able to: PC13. Estimate the cost of all the materials and logistics that will be required to conduct the research successfully within the stipulated time. PC14. Plan a budget that reveals whether the costs of the proposed research are reasonable.

Publication of Research Findings	To be competent, the individual must be able to: PC15. Identify a suitable publisher with thematic areas of interest, and with good research impact factor. PC16. Asses the cost of publishing the article taking into account Article Publishing Cost (APC). PC17. Submit the article as per journal instructions.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of standard procedures guiding research)	The individual on the job must demonstrate knowledge and understanding of: OK 1. National and University Research Ethics Committees. OK 2. Confidentiality of the participants in the research. OK 3. Publishing research findings to add to the body of knowledge and inform policy makers.
B. Technical Knowledge	The individual on the job must demonstrate technical knowledge and understanding of: TK1. Computer software especially Word, PowerPoint, Excel, and use of other statistical packages.
C. Regulatory Knowledge (Knowledge of rules and regulation)	The individual on the job must demonstrate knowledge and understanding of: RK1. Relevant Zambia National Health Research Authority (NHRA) laws and regulations. RK2. Relevant health research Acts.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English language proficiently, and be able to write complex instructions in reports and SOP. CS2. Prepare and provide clear and detailed instructions, details of pathways and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English proficiently, and be able to give detailed instructions. CS4. Read and interpret complex sketches, operating manual drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
B. Professional Skills	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers. CS6. Convey information clearly and concisely to co-workers.
	Decision-Making
	The individual on the job must be able to: PS1. Decide whether the research will contribute significantly to society by informing policy makers of the research findings.

	Plan and Organise
	The individual on the job should be able to: PS2. Decide on how the research will be carried out, and organise how the findings will be communicated to the stakeholders.
	Customer Centricity
	The individual on the job should be able to: PS3. Manage relationships with participants in the research and other clients by ensuring that confidentiality is maintained.
	Problem-Solving
	The individual on the job should be able to: PS 4. Ensure all the equipment and logistics needed to conduct the research are available before beginning the research.
	Analytical Thinking
	The individual on the job should be able to: PS5. Analyse and categorise the challenges that affected the process of carrying out the research.
	Critical Thinking
	The individual on the job should be able to: PS6. Identify the challenges encountered when conducting research, and offer recommendations on how to overcome such challenges in future.

UNIT 6 [This Unit describes the skills and knowledge required by a Biomedical Scientist to play a role in the development and manufacturing of new laboratory equipment, test kits, drugs, and reagents]

Unit No.	06
Unit Title	Development and Manufacturing of New Laboratory Equipment, Test Kits, Drugs, and Reagents
Description	This Unit describes the skills and knowledge required by a Biomedical Scientist to play a role in the development and manufacturing of new laboratory equipment, test kits, drugs, and reagents.
Scope	This Unit covers the following: Advances in Technology and Medicine for Improving Healthcare.
Element	Performance Criteria (PC)
Advances in Technology and Medicine for Improving Healthcare	To be competent, the individual must be able to: PC1. Play a role in developing or recognising new ideas in biomedical sciences that lead to advanced medical technology. PC2. Think flexibly and view current medical technology in different perspectives. PC3. Use imagination or original ideas to contribute to the development of new technology in biomedical sciences and medicine in general. PC4. Collaborate with other experts, including: Engineers, Information and Technology experts, Mathematicians, Biologists and Chemistry experts, in developing new medical technology.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the standard procedures guiding research)	The individual on the job must demonstrate knowledge and understanding of: OK1. Organisational policies, global manufacturing regulations of medical equipment, test kits, drugs, and reagents, and best practices. OK2. Occupational Safety, Health, and Environmental (OSHE) rules, and the organisational regulations and guidelines.
B. Technical Knowledge	The individual on the job must demonstrate technical knowledge and understanding of: TK1. Engineering principles to contribute to the design of new medical equipment, test kits, drugs, and reagents. TK2. Work with other scientists to research how engineering principles apply to biological systems. TK3. Suggest technical alterations to the medical device, test kit, drug or reagent under development during the manufacturing process with other experts.
C. Regulatory Knowledge (Knowledge of	The individual on the job must demonstrate knowledge and understanding of: RK1. Relevant Zambia National Health Research Authority (NHRA) laws.

rules and regulation)	RK2. Relevant health research laws and regulations. RK3. International and national related laws, regulations, and standards regarding new technology in medicine.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in the English language proficiently, and be able to write complex instructions in reports and SOP. CS2. Prepare and provide clear and detailed instructions, details of pathways and sketches to co-workers.
	Reading Skills
	The individual on the job must be able to: CS3. Read English language proficiently, and be able to give detailed instructions. CS4. Read and interpret complex sketches, operating manual drawings or instructions provided for the required work.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS5. Listen attentively and interpret communication/ instructions from the supervisor and other co-workers. CS6. Convey information clearly and concisely to co-workers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Decide whether the medical device, test kit, drug or reagent developed will contribute significantly to society by informing policy makers.
	Plan and Organise
	The individual on the job should be able to: PS2. Collaborate with other experts on how the medical device, test kit, drug or reagent will be developed and manufactured, and communicate such information to the stakeholders.
	Customer Centricity
	The individual on the job should be able to: PS3. Follow the code of conduct relating to the development of new medical device, test kits, drugs or reagent. PS4. Demonstrate that new medical technology will only be used on people when all safety tests have been satisfied.
	Problem-Solving
	The individual on the job should be able to: PS5. Ensure that all the other experts required to develop the new medical device, test kits, drug or reagent are available before beginning the work.

	Analytical Thinking
	The individual on the job should be able to:
	PS6. Analyse and categorise the challenges that affected the process of developing and manufacturing the new medical device, test kits, drug or reagent.
	Critical Thinking
	The individual on the job should be able to:
	PS7. Identify the challenges encountered when contributing to the advancement of medical technology, and offer recommendations on how to overcome such challenges in future.

UNIT 7 [This Unit describes the skills and knowledge required by a Biomedical Scientist to identify areas of entrepreneurship in biomedical sciences]

Unit No.	07
Unit Title	Entrepreneurship Opportunities in Biomedical Sciences
Description	This Unit describes the skills and knowledge required by a Biomedical Scientist to identify areas of entrepreneurship in biomedical sciences.
Scope	This Unit covers the following: Business Opportunities in Biomedical Sciences (for the manufacturing and sale of laboratory equipment, reagents, and commodities; setting up private laboratory practices, and training of new scientists in universities and colleges).
Element	Performance Criteria (PC)
Business Opportunities in Biomedical Sciences	To be competent, the individual must be able to: PC1. Identify and explore the wide range of business opportunities available in the field of biomedical sciences. PC2. Develop, organise, and run a new business. PC3. Act on the business idea to disrupt the current market with a new product or service. PC4. Assess the market and demand for the product or service, and the financial risk associated with the business planned.
Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the standard procedures guiding business set up in Zambia)	The individual on the job must demonstrate knowledge and understanding of: OK1. Relevant national standards, policies, and procedures followed in the registration of a business/company. OK2. Business/company roles, responsibilities, accountabilities and authorities.
B. Technical Knowledge	The individual on the job must demonstrate technical knowledge and understanding of: TK1. Creating a business plan. TK2. Acquiring resources and financing for the new business. TK3. Hiring suitable labour for the business. TK4. Providing leadership and management for the business.
C. Regulatory Knowledge (Knowledge of rules and regulation)	The individual on the job must demonstrate knowledge and understanding of: RK1. Patents and Companies Registration Agency (PACRA). RK2. Zambia Revenue Authority (ZRA). RK3. Zambia Development Agency (ZDA). RK4. Competition and Consumer Protection Commission (CCPC).

Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Issue instructions, recommendations, and commendations in writing. CS2. Conduct performance assessments and develop performance reports.
	Reading Skills
	The individual on the job must be able to: CS3. Read and understand leadership courses.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS4. Manage meetings and discussions. CS5. Provide feedback on Strength Weaknesses, Opportunities, and Threats (SWOT) for the business. CS6. Give instructions to the team. CS7. Listen attentively and comprehend information given by the speakers.
B. Professional Skills	Decision-Making
	The individual on the job must be able to: PS1. Decide whether the business plan will contribute significantly to society by improving healthcare service delivery, while assessing the financial risks related to the business.
	Plan and Organise
	The individual on the job must be able to: PS2. Plan, organise, lead, and control business activities. PS3. Use the SWOT analysis in their functional area of responsibility.
	Customer Centricity
	The individual on the job should be able to: PS4. Follow code of conduct relating to the setting up of the business, and ensure that the product or service offered meets the set national standards and customer expectations.
	Problem-Solving
	The individual on the job must be able to: PS5. Solve complex problems diligently within the business. PS6. Identify problems, apply appropriate problem-solving techniques, and be assertive in decision-making. PS7. Consult widely and identify possible remedies.
	Analytical Thinking
	The individual on the job should be able to: PS8. Analyse and categorise the challenges that affected the setting up of the business plan.

	PS9. Solve problems quickly and effectively using a methodical step-by-step approach to thinking, and break down complex problems into single and manageable components.
	Critical Thinking
	The individual on the job must be able to: PS10. Use common sense and make judgments in day-to-day activities. PS11. Use reasoning skills to identify and resolve basic problems. PS12. Use intuition to identify any potential problems that could arise during business operations.

UNIT 8 [This Unit covers the skills and knowledge required by a Biomedical Scientist in demonstrating leadership and management in the workplace]

Unit No.	08
Unit Title	Leadership and Management in the Workplace
Description	This Unit describes the skills and knowledge required by a Biomedical scientist in Leading and Managing subordinates in the workplace.
Scope	This unit covers the following: • Communicate Effectively in the Workplace. • Carry out Basic Management and Leadership Functions of Planning, Organising, Staffing, Leading, and Controlling. • Contribute to Team and Self-development.
Performance Criteria (PC) with respect to the Scope	
Element	Performance Criteria (PC)
Communicate Effectively in the Workplace	To be competent, the individual must be able to: PC1. Describe the importance of team-based activities, and clearly highlight the key responsibilities they have as team members. PC2. Identify internal and external stakeholders, and meeting their expectations. PC3. Apply available and appropriate feedback mechanisms. PC4. Understand the communication channels and the associated hierarchies.
Carry out Basic Management and Leadership Functions of Planning, Organising, Staffing, Leading, and Controlling	To be competent, the individual must be able to: PC5. Decide on the plans, and take the necessary steps to achieve the objectives. PC6. Delegate tasks and allocate resources to individuals. PC7. Determine the manpower requirements in the working area, and decide their placement. PC8. Motivate and lead the staff for timely achievements of the goals. PC9. Regularly monitor the progress of work in line with the agreed objectives or targets. PC10. Collaborate with HR to be able to conduct training and awareness on code of conduct, and the company's grievances procedure. PC11. Use interpersonal skills to motivate staff to enhance performance in line with set targets.
Contribute to Team and Self-development	To be competent, the individual must be able to: PC12. Describe self-management practices and how they apply to overall team activities in the workplace. PC13. Describe the importance of initiating feedback as a team member towards the overall team development. PC14. Describe the importance of sharing knowledge and experiences for the sake of team development. PC15. Conduct a Gap Analysis to determine training needs for all subordinates.

Knowledge and Understanding (K)	
A. Organisational Context (Knowledge of the whole organisation and its processes)	The individual on the job must be able to demonstrate knowledge and understanding of the: OK1. Company's communication procedures and organogram. OK2. Modes of communication and their associated restrictions if any. OK3. Company's code of ethics.
B. Technical Knowledge	The individual on the job must demonstrate knowledge and understanding of: TK1. How to use computers and other laboratory equipment. TK2. Testing samples to know the underlying problem in a patient. TK3. Maintaining accurate data and records, and producing medical reports. TK4. Conflict and problem-solving skills.
C. Regulatory Knowledge (Knowledge of rules and regulation)	The individual on the job must demonstrate knowledge and understanding of relevant: RK1. Occupational Health and Safety laws. RK2. Industrial and Labour Relations laws. RK3. Workers Compensation Act. RK4. Employment regulations.
Skills (S)	
A. Core Skills/ Generic Skills	Writing Skills
	The individual on the job must be able to: CS1. Write in English language proficiently, and be able to write complex instructions in reports and SOP. CS2. Prepare and provide clear and detailed instructions, details of pathways and sketches to co-workers. CS3. Write concise and detailed notes during an experiment. CS4. Write accurate observations during an experiment. CS5. Write reports into research papers for publication.
	Reading Skills
	The individual on the job must be able to: CS6. Read English language proficiently, and be able to give detailed instructions. CS7. Read and interpret complex, sketches, operating manuals, drawings or instructions provided for the required work and pathology reports.
	Oral Communication (Listening and Speaking Skills)
	The individual on the job must be able to: CS8. Listen attentively and interpret communication and instructions from the supervisor and other co-workers. CS9. Convey information clearly and concisely to co-workers.

B. Professional Skills	Decision-Making
	The Individual on the job holder must be able to: PS1. Collect feedback from team members to ideate a new team workflow. PS2. Research market trends to understand how they will impact company strategy and overall direction of the laboratory. PS3. Learn from mistakes and use data to develop better solutions. PS4. Analyse situations based on facts and data, not assumptions or emotions.
	Plan and Organise
	The individual on the job must be able to: PS5. Plan and organise what, when, who, and how to be communicated based on company communication guidelines. PS6. Divide goals into smaller units and setting deadlines for them. PS7. Decide when to do each task on a project based on workload and deadlines
	Customer Centricity
	The individual on the job must be able to: PS8. Possess deep understanding of people, their joys, their struggles, their lives, their environments, their neighbourhoods, and all of their contexts. PS9. Understand customer problems in order to solve them. PS10. Exhibit servant leadership.
	Problem-Solving
	The individual on the job must be able to: PS11. Active participation in scheduled team activities, rendering contributions in problem-solving and overall decisions of the team PS12. Break down big problems into manageable ones. PS13. Be persistent PS14. Ask for help in times of uncertainty.
	Analytical Thinking
	The individual on the job must be able to: PS15. Brainstorm creative solutions and narrow down to most logical ones. PS16. Gather information through testing and observation. PS17. Develop solutions or deepening understanding of the topic. PS18. Design learning and development programmes that meet the company's needs. PS19. Quantification and forecasting.

	Critical Thinking
	The individual on the job must be able to: PS20. Make solid arguments. PS21. Use reasoning skills to come to logical conclusions. PS22. Identify biases.

5. EQUIPMENT, TOOLS, AND CONSUMABLE MATERIALS

These include, but not limited to:

Equipment and Tools: Haematology Blood Analyser; Chemistry analyser, Flow cytometer; Centrifuges Tissue Processers and Strainers, Incubators, Autoclaves, Microscopes, Blood Culture Machines, Water Bath, PCR Machines, ELISA Machine, HPLS Systems, Mass Spectrometers, CO₂ Incubators, Cell Counters, Microplate Readers, PH Meters, Blood Gas Analysers, Spectrophotometry, fridges, freezers, ovens, safety cabinets and biological safety cabinets, scales and balances, hot plates, etc.

Consumables: Pipettes, gloves, cotton wool, methylated spirits, coverslips, microscope slides, specimen containers, centrifuge tubes, filter papers, masks, plastic disposable cuvettes, lancets, cryovials, capillary tubes, microtomes, cell culture dishes and plates, syringes, phlebotomy needles and tubes, reagent bottles, staining solutions, filter tips, paraffin wax for tissue embedding, swabs, microbiology culture media, pipette tips, etc.

6. JOBHOLDER DILEMMAS/CHALLENGES AND COMPLEXITIES

Biomedical Scientists face challenges such as: obsolete and/or inappropriate equipment and tools; budgetary constraints; inadequate product costing skills; poor technical skill base; bureaucracy in procurement procedures; lack of appreciation of preventive maintenance by non-scientist management staff; labour intensive nature of the work; rapid change of technology and materials; lack of personal protective equipment; climate change; cyber warfare; inconsistencies in company and government policies and regulations, etc.

6.1 Alternative Choices (Solutions) to Dilemmas and Complexities

Solutions to challenges include: selecting and procuring appropriate equipment and tools for the job; supporting capacity building through training; identifying and utilising suitable adaptation and mitigation measures against the effects of climate change; utilising appropriate cyber security measures to protect against cyber warfare; inclusion of biomedical science professionals in management teams; deployment of automation where feasible; provision of personal protective equipment; participate in lobbying and formulation of policies, allocation of adequate financial resources, etc.

7. WORKING CONDITIONS/ENVIRONMENT

Biomedical Scientists work with a variety of machinery, toxic substances, and volatile, radioactive, infective, and many other dangerous materials. Their work environment is susceptible to fires, explosions, structural failures, and equipment malfunctions. Working conditions include: hot conditions where air conditioning is absent; standing/walking for long hours; lifting materials; working in day or night shifts; areas with limited lighting or ventilation, etc.

8. PARTIES INVOLVED/INTERACTING WITH THE JOBHOLDER OR TRAINEE

8.1 Internal/Within the Organisation

Management, supervisors, subordinates, and other section members, etc.

8.2 External/Outside the Organisation

Government regulators, professional bodies, fellow Biomedical Scientists from other companies, labour unions, clients, suppliers, students/interns, etc.

9. PHYSICAL DEMANDS ON THE BODY

- Physique to sustain strenuous conditions;
- Be able to walk and stand for long periods of time;
- Bend, stretch, twist, or reach out;
- Be able to lift relatively heavy materials, tools and equipment;
- Be able to use fingers, hands, and feet with ease to complete the assigned task (dexterity);
- Etc.

ANNEX A

Criteria for Assessments based on this NOS

A.1 Guidelines for Assessment

A.1.1 Criteria for assessment for curricula and learning programmes based on this NOS will be created by curricula and programme developers. Each Performance Criteria (PC) will be assigned marks proportional to its importance in the NOS. Curricula and programme developers will also lay down proportion of marks for theory and practical skills for each performance criteria, giving more weight to practical skills.

There shall be allocated the 'Total Mark', which will be the sum of all marks in each Unit, distributed across the number of PCs in that particular Unit. The 'Out of' mark will be the mark allocated to each PC, which will be shared between theory and skills practical assessments.

A.1.2 Individual awarding/assessment bodies or institutions and other users of the NOS will create unique question papers for the theory part, and evaluations for skill practical part for their respective candidates.

ANNEX B

NOS Version Control

This Annex gives details necessary for the tracking of the NOS versions based on the number of revisions.

NOS Code	NOS.BS.01		
ZQF Level	7	Version Number	01
Sector	Manufacturing	Date of Approval	December 2023
Sub Sectors	NGO, Research, Academia, Manufacturing Regulatory	Date of Last Review	N/A
Occupation	Biomedical Science	Date of Next Review	December 2028



*“You learn,
We Standardise”*

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